

```
drwxr-xr-x. 2 root root 4096 Mar 3 17:35 Desktop
```

Figure 1: Detailed listing of contents with permissions

first column of the output contains various permissions as explained below:

r w x	r - x	r - x
User owner	Group owner	Other users

How permissions are calculated

The decimal value of *read* is 4, of *write* is 2, of *execute* is 1, and of 'No permission' is 0.

So, if we say that a file has *read* and *execute* permissions, it means the permission is $4+0+1=5$.

A **rwxr-xr--** permission can be read as $(4+2+1)(4+0+1)(4+0+0)=754$, where 7 is implemented for the user owner, 5 for the group owner and 4 for all the other users.

We can also understand this by using binary-to-decimal conversion:

Permissions:	r	w	x
Binary bits:	1	0	1
Conversion:	$1*2^2 + 0*2^1 + 1*2^0$ $4+0+1=5$		

A file with permissions **777** will give all the rights to everyone using your computer.



Note: If you want your file to be readable and executable by the owner user, only readable by group members and offer no access to any other users, you should set permissions as 540.

Managing owner user, group owner and other user permissions

Now, the question is how do we change the permissions for implementing security? The answer is the **chmod** command.

Let's suppose we create a file named **f_shobhit** using the command **touch f_shobhit**. The default permissions attained by this file will be **rwx-rwx-rwx**. To change the permissions of the file **f_shobhit** using the numeric mode, execute the following command:

```
chmod 540 f_shobhit
```

Now, the permissions will become **r-xr-----**

Permissions are always assigned from left to right and if no permission is specified, 0 is taken as default.

```
chmod 5 f_shobhit
```

In the above example, the permission's value will be taken as 005, where 5 is assigned to other users and the resulting permissions will become: **-----r-x**.

```
chmod 45 f_shobhit
```

In the above example, the value will be taken as 045 and resulting permissions will be: **--r--r-x**.

Permissions can also be managed using the symbolic mode as shown below:

```
chmod u+rx,g+x,o-x f_shobhit
```

The result of the above command will be: **rwxr-xr-x**. The **+** symbol adds a permission to a specific octet while the **-** symbol takes away permissions. **u** represents the owner user octet, **g** represents the group owner octet, **o** represents the other users octet, and **a** collectively represents all octets.

```
chmod a=r f_shobhit
```

The above command will assign *Read only* permission to all the octets at once.

Effects of permissions on files

By default, the user name is the user owner and owner group of a file's contents. In the above examples, the user **shobhit** was logged in and created the file **f_shobhit**, so automatically **shobhit** becomes the user owner and owner group. To discuss the effects of permissions, file **f_shobhit** is moved to a directory **/home/Common** such that permissions **777** are implemented on the **Common** directory. This is done so that all the members of the group and other users can have access to the file.

Let us clear all the permissions from the file (leaving the right side of the permission assignment expression blank will do the trick). Then the step-by-step effects of permissions will be elaborated by assigning one permission at a time.

```
chmod a= f_shobhit
```

1. Now, execute the following command:

```
cat f_shobhit
```

The user owner will not be able to use **cat**, **vim** and other commands on a file. Instead, the user will get a 'Permission Denied' error (see Figure 2). Although **vim** is meant for editing a file, due to no-read permissions, the contents will not be read by **vim** and so can't be modified. Similarly, neither members of group **shobhit** nor other users will be able to read the file.

2. Now, add *read permission* to user owner:

```
chmod u+r f_shobhit
```

The **cat f_shobhit** command will now work perfectly for the user owner, but members of the group **shobhit** and other users still can't read the file.